

Outdoor WiFi + Camera Installation

Sand Volleyball Courts - Design & Placement Report

Manor Lanes / WNY Social Sports - 150 Grand Island Blvd

Rev. 2 - UDB-Switch (wireless-uplink) architecture

Site	3 outdoor sand volleyball courts + 2 patios (P1, P2)
Platform	Ubiquiti UniFi (Network + Protect)
Installer	Brian Russo (owner-installer)
Controller / NVR	CloudKey+ (SSD) + Network Video Recorder Pro + 16 TB HDD (existing)
Key constraints	No trenching; no new cable across sand/drainage; cold-weather (Buffalo NY)
Backhaul method	UniFi UDB-Switch: WiFi-7 wireless-uplink switches at each pole (no outdoor cable)
Access points	3x U6 Mesh Pro - relocate 2 off building wall + add 1 new
Net new hardware	approx. \$1,500 - \$2,000 (most network gear repositioned, not repurchased)

Design summary

Each court pole carries a UniFi UDB-Switch - a WiFi-7 switch with a built-in wireless uplink. It backhauls to the UniFi WiFi over the air (no cable across the sand or drainage) AND supplies local PoE to that pole's cameras and access point. A U6 Mesh Pro re-mounted on the building (off the concrete wall) is the wired uplink anchor; the center-pole UDB-Switch links to it, and the Court 3 UDB-Switch links (relay) to the center-pole AP. The three U6 Mesh Pro APs are repositioned from the failed concrete-wall mounting onto poles, where coverage radiates over the sand. Cameras: 3 fixed bullets (one per court); the AI PTZ and a G6 180 remain on the parking lot.

1. Background & Existing Conditions

Problem identified

- Two U6 Mesh Pro APs were wall-mounted on the concrete back of the building.
- Heat-map survey showed strong signal (-30 to -50 dBm) only at the wall; the entire court surface measured -70 to -80 dBm (marginal-to-unusable), failing under league crowds.
- Root cause: omnidirectional APs flat on concrete radiate left/right along the wall and reflect/absorb into the concrete - signal does not project across the open sand.

Fix

- Reposition the U6 Mesh Pro APs off the wall onto field poles so coverage radiates over the sand.
- Use UDB-Switch wireless-uplink switches for backhaul - no cable across the sand/drainage; each pole gets local PoE for its cameras + AP.

Existing UniFi equipment to be REUSED (no purchase)

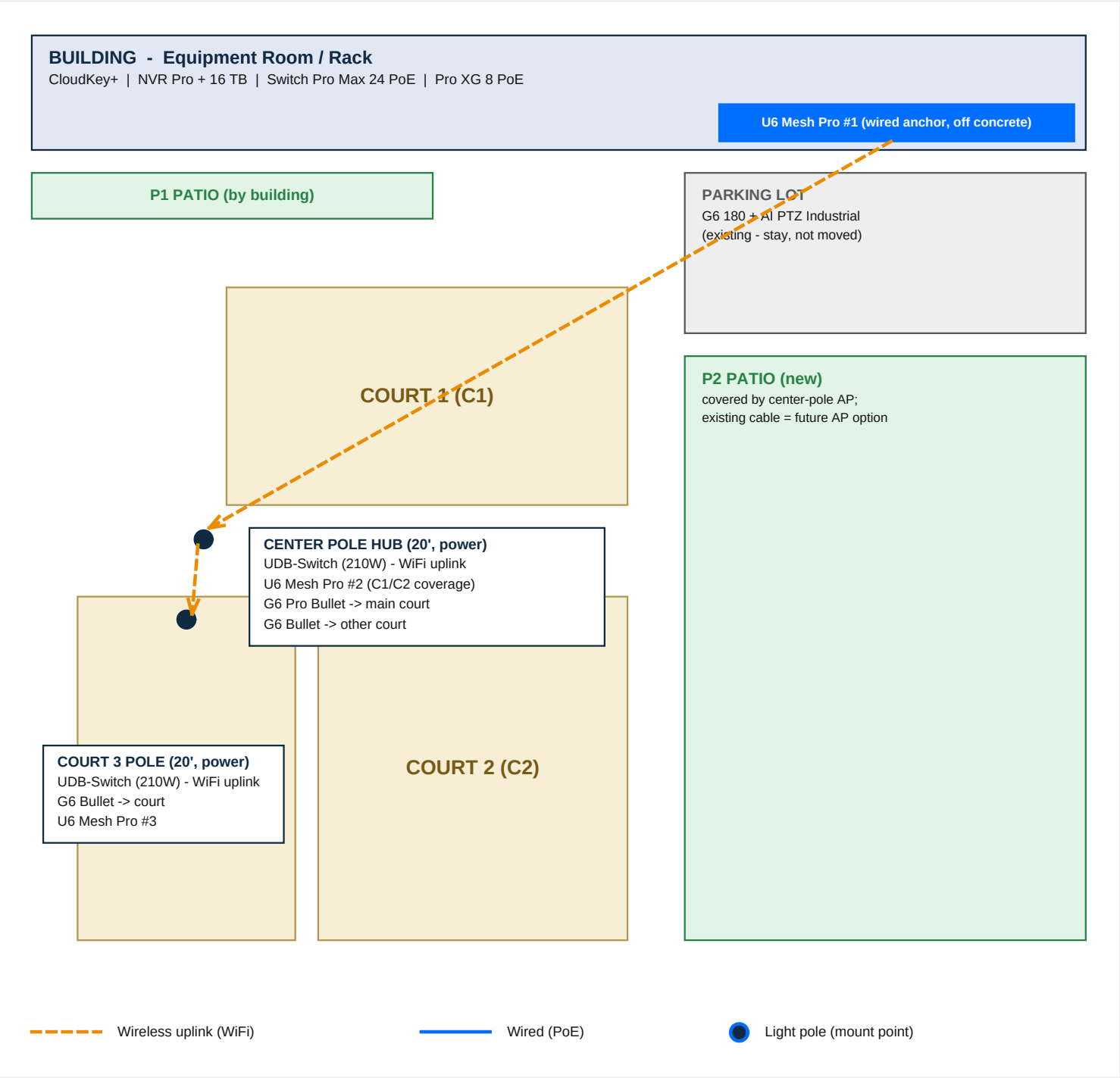
Item	Qty	Role in this project
Network Video Recorder Pro + 16 TB HDD	1	Records all court cameras
CloudKey+ (SSD)	1	UniFi Network/Protect controller
Switch Pro Max 24 PoE / Pro XG 8 PoE	2 / 1	Rack; powers wired building AP
Access Point U6 Mesh Pro	3*	*Relocate 2 off wall + add 1 new = 3 on poles
AI PTZ Industrial (IP66, -40C)	1	STAYS on parking lot (not moved)
Camera G6 180	1	STAYS on parking lot (do NOT use for courts)
Switch Flex 2.5G PoE + 210W adapter	1	Not needed at poles (UDB-Switch replaces it)

Confirmed site facts

- Center pole: new 20 ft pole (2 lights) between Court 1 and Court 2 - POWER PRESENT.
- Court 3 pole: 20 ft light pole - POWER PRESENT.
- P2 patio post: existing Cat6 pulled to it but NO power (kept as a future AP option only).
- Rack is inside the building, ~130 ft from the center-pole location.
- Courts each ~29'6" x 59'1" (approx 30 x 60 ft); perimeter light poles ~20 ft.
- Backhaul is wireless: a WiFi-7 UDB-Switch uplinking to a WiFi-6 U6 Mesh Pro links at WiFi-6 rates (hundreds of Mbps)
- far above the ~50-80 Mbps the cameras need.

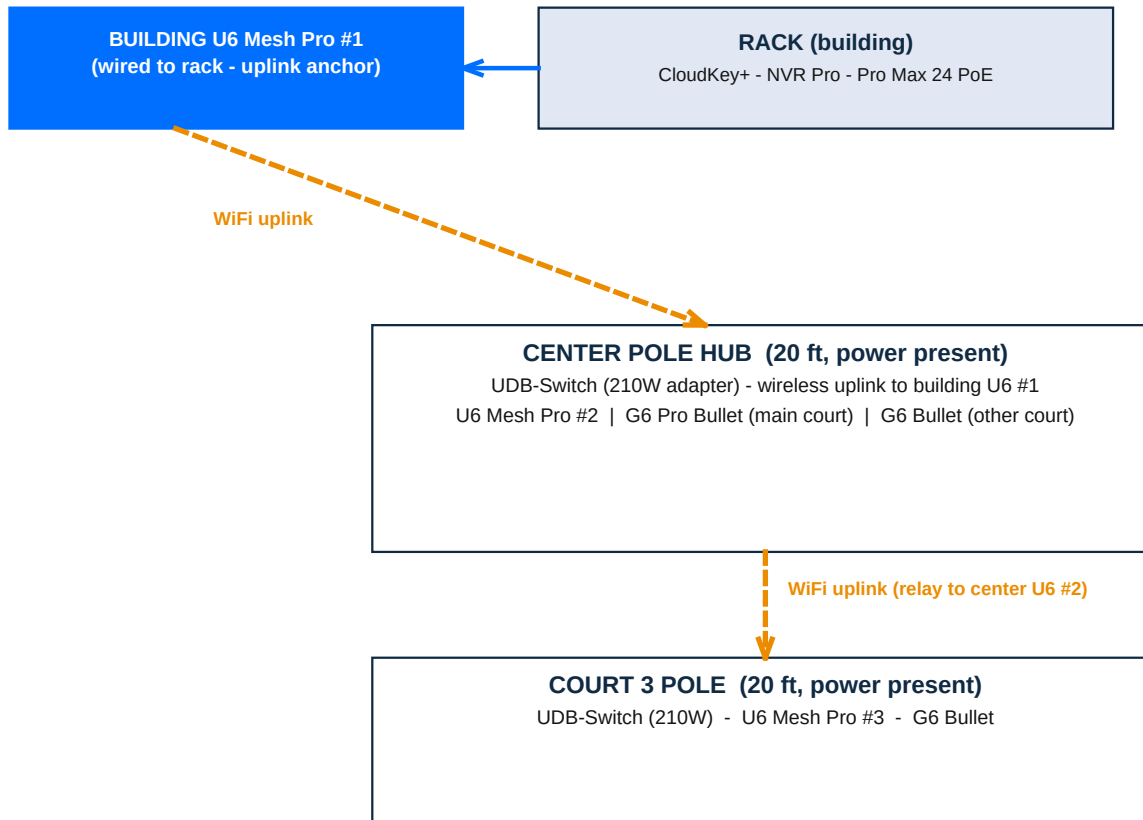
2. Site Layout & Device Placement

Schematic - relative positions per site photos/CAD; not to exact scale. Solid = wired (PoE). Dashed = wireless uplink (WiFi).



Orientation per owner: building at top with rack; P1 patio adjacent to building; Court 1 nearer building, Court 2 further; Court 3 to the left behind Court 2; P2 (new patio) along the right.

3. Network Topology & Wireless Uplinks



How the wireless backhaul works

- The UDB-Switch is a WiFi-7 switch with a built-in wireless uplink: it joins the UniFi WiFi over the air and provides PoE locally - so no cable runs out to the poles.
- Center pole UDB-Switch uplinks to the building U6 Mesh Pro #1 (~130 ft, line-of-sight).
- Court 3 UDB-Switch uplinks (relay) to the center-pole U6 Mesh Pro #2 (Court 3 sits behind Court 2, so it chains through the elevated center pole).
- WiFi-7 uplink to a WiFi-6 AP runs at WiFi-6 rates - hundreds of Mbps, far above the ~50-80 Mbps needed for 3x 4K cameras + AP clients.
- Each UDB-Switch is powered by a 210W adapter (up to 185W PoE) - ample for cameras + AP.
- Position the building U6 with clear line-of-sight to the center pole (mast/standoff off the concrete) so the uplink is strong.

4. Device-by-Device Placement & Mounting

Location	Device	Placement / Aim	Height	Power & Data
Building	U6 Mesh Pro #1 (relocated)	Mast/standoff OFF concrete; LoS to center pole	Roofline	Wired PoE from rack switch
Center pole	UDB-Switch (in NEMA enclosure)	Wireless uplink to building U6 #1	~6 ft (reach)	AC from pole light circuit + 210W adapter
Center pole	G6 Pro Bullet	Aim down MAIN court (zoom set to fill court)	~18-20 ft	PoE from UDB-Switch
Center pole	G6 Bullet	Aim down the other court	~18-20 ft	PoE from UDB-Switch
Center pole	U6 Mesh Pro #2 (relocated)	Radiate over C1/C2; reaches P1/P2	~15-18 ft	PoE from UDB-Switch
Court 3 pole	UDB-Switch (in NEMA enclosure)	Wireless uplink to center-pole U6 #2	~6 ft (reach)	AC from pole light circuit + 210W adapter
Court 3 pole	G6 Bullet	Aim down length of Court 3	~18-20 ft	PoE from UDB-Switch
Court 3 pole	U6 Mesh Pro #3 (new)	Radiate over Court 3	~15-18 ft	PoE from UDB-Switch
Parking lot	G6 180 + AI PTZ Industrial	No change - stay on lot	existing	existing

Camera aiming: mount each bullet at one END of its court and aim down the 60 ft length for full coverage and best pixel density. The G6 Pro Bullet's optical zoom lets you fill the frame with the main court (best detail + low light) and doubles as the special-games detail cam. Do not use the G6 180 for court detail.

5. Bill of Materials

New hardware to purchase

Item (UniFi SKU)	Qty	Unit	Line
Device Bridge Switch (UDB-Switch-US)	2	\$299	\$598
AC Adapter 210W (UACC-Adapter-AC-210W)	2	\$79	\$158
Access Point U6 Mesh Pro (new; + relocate 2 owned)	1	\$179	\$179
Camera G6 Pro Bullet (UVC-G6-Pro-Bullet-B) - main court	1	\$479	\$479
Camera G6 Bullet (UVC-G6-Bullet-B) - other 2 courts	2	\$199	\$398
Outdoor NEMA enclosure sized for UDB-Switch	2	~\$90	~\$180
Ethernet Surge Protection Outdoor (UACC-ETH-SP-Pro)	2-4	\$19	\$38-76
Pole mounts / building mast + short patch cables	1 lot	~\$120	~\$120

Estimated net new spend: approx. \$2,150 - \$2,200

Camera option: substitute 3x G6 Bullet (~\$597 total) for the 1 Pro + 2 regular mix (~\$877) if budget-driven - regular bullets give good full-court coverage; the Pro adds zoom detail and low-light on the main court. Note: tariff/memory surcharges from the UniFi store are extra.

Removed from earlier cart (not needed)

- U7 Outdoor APs x2 - not needed; U6 Mesh Pro is already omnidirectional and outdoor-rated.
- Switch Flex Utility x1 - that is an IP65 enclosure for a Flex SWITCH, not the larger UDB-Switch; use UDB-sized NEMA enclosures instead.
- Extra 2 cameras (cart had 5; design needs 3).

Reused (already owned - \$0)

- 3x U6 Mesh Pro APs (relocate 2 off building wall, add the 1 new = 3 on poles)
- Network Video Recorder Pro + 16 TB HDD - recording
- CloudKey+ controller; rack PoE switches power the wired building AP
- AI PTZ Industrial + G6 180 - stay on parking lot

6. Installation & Configuration

Physical install sequence

1. Re-mount a U6 Mesh Pro on the building OFF the concrete (short mast/standoff at the roofline) with clear line-of-sight to the center pole; wire it to a rack PoE port. This is the uplink anchor.
2. Mount a NEMA enclosure on the center pole; install the UDB-Switch + 210W adapter; have an electrician land pole-light-circuit power to it (no trenching - power is at the pole).
3. Mount center-pole devices: G6 Pro Bullet (main court), G6 Bullet (other court), U6 Mesh Pro #2. Patch all to the UDB-Switch PoE ports.
4. Repeat at Court 3: NEMA enclosure + UDB-Switch + 210W adapter (pole power), G6 Bullet, U6 Mesh Pro #3.
5. Install ETH-SP surge protectors on the wired building-AP run and any cable entering the building.
6. Aim cameras down each court; tilt APs slightly down toward the sand.

UniFi configuration

- Adopt the building U6 (wired) first; build one outdoor WLAN; enable band steering.
- Adopt each UDB-Switch and set its wireless uplink to the correct AP (center -> building U6; Court 3 -> center U6). Confirm uplink RSSI better than -60 dBm and a healthy link rate.
- Adopt the pole APs (U6 #2, #3). Set fixed, non-overlapping 5 GHz channels across the 3 APs; reduce TX power from High to Medium once on poles; verify with a follow-up heat map.
- Adopt cameras in UniFi Protect; confirm recording to NVR Pro; set AI/motion zones to the courts; set the G6 Pro Bullet zoom to fill the main court.
- Confirm each UDB-Switch PoE load is within budget (185W with the 210W adapter).

PoE budget per pole (UDB-Switch, 185W available with 210W adapter)

Pole load	Approx draw
Center pole: G6 Pro + G6 Bullet + U6 Mesh Pro	~32 W
Court 3: G6 Bullet + U6 Mesh Pro	~20 W
Headroom on 185W budget	very large

7. Commissioning & Acceptance Checklist

Wireless backhaul (UDB-Switch uplinks)

- ☐ Center-pole UDB-Switch uplink online; RSSI better than -60 dBm; link rate healthy.
- ☐ Court 3 UDB-Switch uplink (relay) online; all Court 3 devices reachable.
- ☐ Pull a phone/laptop speed test through each pole - well above camera needs.

WiFi coverage (re-survey)

- ☐ Re-run the WiFiman heat map across all 3 courts + both patios.
- ☐ Court surface now reads -55 dBm or better (was -70 to -80).
- ☐ No dead pocket on P1/P2 (existing P2 cable can feed an added AP if needed).
- ☐ Roaming: a phone walks court-to-court without dropping.

Cameras (UniFi Protect)

- ☐ All 3 court bullets online and recording to NVR Pro.
- ☐ Each bullet frames its full court; G6 Pro zoom set to fill the main court.
- ☐ Night check: image usable under the Musco court lights.
- ☐ Retention check: 16 TB HDD holds target days with the added cameras (add HDD if short).

Power & weatherproofing

- ☐ UDB-Switch + 210W adapter sealed inside NEMA enclosures; cable glands tight; drip loops.
- ☐ ETH-SP surge protectors grounded on the wired building run.
- ☐ No PoE over-budget warnings on either UDB-Switch.

Sign-off

Installer: _____

Date: _____

Notes: _____
